

1 Principles of Monte Carlo

1.1 Introduction Monte Carlo integration

Consider estimating $\alpha = \int_0^1 f(x) dx$. If $U \sim \text{Unif}(0, 1)$, then $\alpha = \mathbb{E}[f(U)]$. Given independent samples $U_1, \dots, U_n \sim \text{Unif}(0, 1)$, the Monte Carlo estimator is

$\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n f(U_i)$.

By the strong law of large numbers, $\hat{\alpha}_n \rightarrow \alpha$ a.s. as $n \rightarrow \infty$. By the central limit theorem,

$\hat{\alpha}_n - \alpha \approx N(0, \sigma_f^2/n)$,

where $\sigma_f^2 = \text{var}(f(U))$. Thus the standard error is $\sigma_f/\sqrt{n}$. Halving the standard error requires increasing the number of samples by a factor of 4.

Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is $O(n^{-2})$. The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

1.2 First Example: European Call Option

Risk-neutral valuation

Let $S(t)$ be the stock price at time t . A European call option with strike K and maturity T has payoff $(S(T) - K)^+ = \max\{0, S(T) - K\}$. Under the risk-neutral measure $\mathbb{Q}$, the time-0 price is

$V_0 = \mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (S(T) - K)^+]$,

where r is the continuously compounded risk-free rate.

Stock-price dynamics

Under $\mathbb{Q}$, geometric Brownian motion satisfies $\frac{dS(t)}{S(t)} = r dt + \sigma dW(t)$, where $W(t)$ is a standard Brownian motion. The exact solution at time T is

$S(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma W(T)\right)$.

Since $W(T) = \sqrt{T}Z$ with $Z \sim N(0, 1)$,

$S(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\right)$.

Simulation algorithm

For $i = 1, \dots, n$, generate $Z_i \sim N(0, 1)$, then set

$S_i(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\right)$,

$C_i = e^{-rT} (S_i(T) - K)^+$.

The Monte Carlo estimator is $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$. It is unbiased, $\mathbb{E}[\hat{C}_n] = C$, and strongly consistent, $\hat{C}_n \rightarrow C$ with probability 1.

Confidence interval

The sample standard deviation of the simulated discounted payoffs is

$s_C = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (C_i - \hat{C}_n)^2}$.

An asymptotic $1 - \delta$ confidence interval is

$\hat{C}_n \pm z_{\delta/2} \frac{s_C}{\sqrt{n}}$.

Since s_C is estimated, one may replace the normal quantile with the corresponding t -quantile with $n - 1$ degrees of freedom.

Black-Scholes benchmark

The Black-Scholes value of a European call is

$BS(S, \sigma, T, r, K) = S\Phi(d_1) - e^{-rT}K\Phi(d_2)$,

where

$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$,

$d_2 = \frac{\log(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.

1.3 Simulation Layers

Layered construction

Monte Carlo derivative pricing is built in layers:

$U_1 \rightarrow Z_1 \rightarrow S_1(T)$ or $\{S_1(t_j)\}_{j=1}^n \rightarrow C_1 \rightarrow \hat{C}_n, s_C$.

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

1.4 Second Example: Asian Option

Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff $(\bar{S} - K)^+$, where

$\bar{S} = \frac{1}{m} \sum_{i=1}^m S(t_j)$,

and $0 = t_0 < t_1 < \dots < t_m = T$. Its time-0 value is

$\mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (\bar{S} - K)^+]$.

To estimate this value, simulate the path values $S(t_1), \dots, S(t_m)$, compute the average $\bar{S}$, discount the payoff, and average over independent replications.

1.5 Efficiency of Simulation Estimators

Central limit theorem

Suppose $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$, where C_i are i.i.d., $\mathbb{E}_0^{\mathbb{Q}}[C_i] = C$, and $\text{var}(C_i) = \sigma_C^2 < \infty$. Then

$\frac{\hat{C}_n - C}{\sigma_C/\sqrt{n}} \Rightarrow N(0, 1)$,

or equivalently

$\sqrt{n}(\hat{C}_n - C) \Rightarrow N(0, \sigma_C^2)$.

Hence $\hat{C}_n - C \approx N(0, \sigma_C^2/n)$. For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

1.6 Bias

General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite n , but become asymptotically unbiased as computational effort increases.

Ratio estimator

For i.i.d. samples (X_i, Y_i) , the estimator $\bar{X}/\bar{Y}$ for $\mathbb{E}[X]/\mathbb{E}[Y]$ is generally biased because

$\mathbb{E}\left[\frac{\bar{X}}{\bar{Y}}\right] \neq \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$.

However,

$\frac{\bar{Y}}{\bar{X}} \rightarrow \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as $n \rightarrow \infty$.

Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

$S((j+1)h) = S(jh) + rS(jh)h + \sigma S(jh)\sqrt{h}Z_{j+1}$,

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step h , at the cost of more transitions per path.

Payoff discretization error

For a continuously averaged Asian option,

$\bar{S} = \frac{1}{T} \int_0^T S(u) du$.

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a smaller simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payoff discretization error comes from approximating the payoff functional.

1.7 Mean Square Error

Bias-variance decomposition

For an estimator $\hat{\alpha}$ of α , the mean square error is

$\text{MSE}(\hat{\alpha}) = \mathbb{E}\left[(\hat{\alpha} - \alpha)^2\right]$.

It decomposes as

$\text{MSE}(\hat{\alpha}) = (\mathbb{E}[\hat{\alpha}] - \alpha)^2 + \mathbb{E}\left[(\hat{\alpha} - \mathbb{E}[\hat{\alpha}])^2\right]$,

or

$\text{MSE}(\hat{\alpha}) = \text{Bias}^2(\hat{\alpha}) + \text{Variance}(\hat{\alpha})$.

Thus MSE balances bias and variance.

2 Generating Random Numbers and Random Variables

2.1 Random Number Generation

Uniform inputs

Monte Carlo simulation is driven by apparently random numbers. Ideally we want $U_1, U_2, \dots$ such that each $U_i \sim \text{Unif}[0, 1]$ and the U_i are mutually independent. Independence is crucial: U_i should not be predictable from $U_1, \dots, U_{i-1}$.

A pseudorandom number generator produces a finite deterministic sequence $u_1, \dots, u_K \in [0, 1]$. A good generator makes short segments hard to distinguish from independent uniforms. Desirable properties: long period, reproducibility, speed, portability, and statistical randomness.

Linear congruential generator

A linear congruential generator is

$x_{i+1} = ax_i \bmod m$, $u_{i+1} = x_{i+1}/m$.

Here a is the multiplier, m the modulus, and $x_0 \in \{1, \dots, m - 1\}$ the seed. The modulo operation is $y \bmod m = y - \lfloor y/m \rfloor m$.

Once a value repeats, the whole sequence repeats. The generator has full period if it produces all $m - 1$ nonzero values before repeating. Large m alone is not enough: poor choices of a, m can produce short cycles.

2.2 General Sampling Methods

Uniform distribution

Assume independent uniforms satisfying $P(U_i \leq u) = 0$ if $u < 0$, $P(U_i \leq u) = u$ if $0 \leq u \leq 1$, and $P(U_i \leq u) = 1$ if $u > 1$. Sampling methods transform these uniforms into variables with desired distributions.

2.3 Inverse Transform Method

General method

To sample X with CDF F , set

$X = F^{-1}(U)$, $U \sim \text{Unif}[0, 1]$.

If F is not strictly increasing, use the generalized inverse

$F^{-1}(u) = \inf\{x : F(x) \geq u\}$.

This chooses the smallest compatible x . Validity follows from

$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$.

Flat sections of F have zero probability: if F is constant on $[a, b]$, then $P(a < X \leq b) = F(b) - F(a) = 0$.

Examples

Exponential with mean θ : $F(x) = 1 - e^{-x/\theta}$ for $x \geq 0$, so

$X = -\theta \log(1 - U)$, equivalently $X = -\theta \log U$.

Rayleigh example:

$F(x) = 1 - e^{-2x(x-b)}$, $x \geq b$,

so the inverse-transform sample is

$X = \frac{b}{2} + \frac{\sqrt{b^2 - 2\log U}}{2}$.

Discrete distribution: For outcomes $c_1 < \dots < c_n$ with probabilities p_i , define $q_i = 0$ and

$q_i = \sum_{j=1}^i p_j$.

Generate U , find K with $q_{K-1} < U \leq q_K$, and set $X = c_K$. Binary search can implement the lookup.

Conditional sampling

To sample X conditional on $a < X \leq b$, with $F(a) < F(b)$, generate

$V = F(a) + [F(b) - F(a)]U$.

Then $F^{-1}(V)$ has the conditional distribution because

$P(F^{-1}(V) \leq x) = \frac{F(x) - F(a)}{F(b) - F(a)}$.

2.4 Acceptance-Rejection Method

General method

To sample from a target density f on $\mathcal{X}$, choose a convenient proposal density g and a constant c such that

$f(x) \leq cg(x)$, $x \in \mathcal{X}$.

Generate $X \sim g$ and $U \sim \text{Unif}[0, 1]$. Accept X if

$U \leq \frac{f(X)}{cg(X)}$;

otherwise reject and repeat. If Y is the accepted value, then $P(Y \in A) = P(X \in A \mid U \leq f(X)/(cg(X)))$.

The acceptance probability is

$P(U \leq f(X)/(cg(X))) = \int_{\mathcal{X}} \frac{f(x)}{cg(x)} g(x) dx = 1/c$.

Hence

$P(Y \in A) = c \int_A \frac{f(x)}{cg(x)} g(x) dx = \int_A f(x) dx$.

Thus Y has density f . The number of proposals until acceptance is geometric with mean c , so smaller c and tighter bounds are better.

Beta distribution

The beta density on $[0, 1]$ is

$f(x) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} x^{\alpha_1-1} (1 - x)^{\alpha_2-1}$.

Here

$B(\alpha_1, \alpha_2) = \int_0^1 x^{\alpha_1-1} (1 - x)^{\alpha_2-1} dx = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}$,

and

$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$, $\text{Re}(z) > 0$.

For integer $n > 0$, $\Gamma(n) = (n - 1)!$. If $\alpha_1, \alpha_2 \geq 1$ and at least one exceeds 1, the mode is $\frac{\alpha_1-1}{\alpha_1+\alpha_2-2}$.

Let c be the maximum value of f and use $g(x) = 1$ on $[0, 1]$. Generate U_1, U_2 until $cU_2 \leq f(U_1)$, then return U_1 .

Normal from double exponential

For the double exponential density $g(x) = e^{-|x|}/2$ and standard normal density

$f(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$,

the ratio satisfies

$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}} e^{-x^2/2+|x|} \leq \sqrt{\frac{2}{\pi}} \approx 1.3155 \approx c$.

The rejection test may be written as

$u > \exp\left(-\frac{1}{2}x^2 + |x| - \frac{1}{2}\right) = \exp\left(-\frac{1}{2}(|x| - 1)^2\right)$.

Using symmetry: generate U_1, U_2, U_3 , set $X = -\log U_1$, reject if $U_2 > \exp[-0.5(X - 1)^2]$, then set $X \leftarrow -X$ if $U_3 \leq 0.5$.

2.5 Normal Random Variables and Vectors

Multivariate normal

A d -dimensional normal is written $N(\mu, \Sigma)$, where μ is a d -vector and Σ is a $d \times d$ covariance matrix. A covariance matrix must be symmetric and positive semidefinite:

$x^\top \Sigma x \geq 0$, $x \in \mathbb{R}^d$.

If Σ is positive definite, the density is

$\phi_{\mu, \Sigma}(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right]$.

If $X \sim N(\mu, \Sigma)$, then $X_i \sim N(\mu_i, \sigma_i^2)$ with $\sigma_i^2 = \Sigma_{ii}$, and

$\text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \Sigma_{ij}$,

$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}$, equivalently $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$.

If Σ is positive semidefinite but singular, define $X = \mu + AZ$ with $Z \sim N(0, I_d)$ and $AA^\top = \Sigma$. Then X has no full d -dimensional density when $\text{rank}(\Sigma) < d$.

Useful properties

Linear transformation: if $X \sim N(\mu, \Sigma)$, then

$AX \sim N(A\mu, A\Sigma A^\top)$.

Conditioning formula: for a partitioned normal vector with $\Sigma_{[22]}$ full rank,

$(X_{[1]} \mid X_{[2]} = x) \sim N(\mu_{[1]} + \Sigma_{[12]}\Sigma_{[22]}^{-1}(x - \mu_{[2]}), \Sigma_{[11]} - \Sigma_{[12]}\Sigma_{[22]}^{-1}\Sigma_{[21])}$.

Moment generating function:

$\mathbb{E}[\exp(\theta^\top X)] = \exp(\mu^\top \theta + \frac{1}{2}\theta^\top \Sigma \theta)$.

2.6 Generating Normals

Box-Muller method

The Box-Muller method generates a bivariate standard normal. If $Z \sim N(0, I_2)$, then $R = Z_1^2 + Z_2^2$ is exponential with mean 2:

$P(R \leq s) = 1 - e^{-s/2}$.

Given $R, (Z_1, Z_2)$ is uniformly distributed on the circle of radius $\sqrt{R}$. Generate

independent U_1, U_2 , set

$R = -2\log U_1$, $V = 2\pi U_2$,

and return

$Z_1 = \sqrt{R} \cos V$, $Z_2 = \sqrt{R} \sin V$.

Multivariate normals

To sample $X \sim N(\mu, \Sigma)$, generate $Z \sim N(0, I_d)$, find A such that

$AA^\top = \Sigma$, and set $X = \mu + AZ$. Then $X \sim N(\mu, \Sigma)$.

2.7 Cholesky Factorization

Positive definite case

Cholesky represents $\Sigma = AA^\top$ with A lower triangular. For

$\Sigma = \begin{pmatrix} \sigma_1^2 & & \\ \sigma_1\sigma_2\rho & \sigma_2^2\rho^2 & \\ \sigma_1\sigma_2\rho & \sigma_2^2\rho^2 & \sigma_2^2 \end{pmatrix}$,

a Cholesky factor is

$A = \begin{pmatrix} \sigma_1 & & 0 \\ \rho\sigma_2 & \sqrt{1-\rho^2}\sigma_2 & \end{pmatrix}$.

Thus

$X_1 = \mu_1 + \sigma_1 Z_1$,

$X_2 = \mu_2 + \sigma_2 \rho Z_1 + \sigma_2 \sqrt{1 - \rho^2} Z_2$.

For general d , use

$\Sigma_{ij} = \sum_{k=1}^i A_{ik} A_{jk}$, $j \leq i$.

The recursive formulas are

3.6 Multi-Asset GBM

Correlated assets

Let X_t be standard Brownian motions with correlations ρ_{ij} . A multidimensional GBM is

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dX_i(t), \quad i = 1, \dots, d.$$

Define $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$. Then $(\sigma_1 X_1, \dots, \sigma_d X_d) \sim \text{BM}(0, \Sigma)$ and $S = (S_1, \dots, S_d) \sim \text{GBM}(\mu, \Sigma)$.

The price covariance is

$$\text{Cov}[S_i(t), S_j(t)] = S_i(0)S_j(0)e^{(\mu_i + \mu_j)t} \left(\rho_{ij} \sigma_i \sigma_j t - 1 \right).$$

If $AA^T = \Sigma$ and a_i is the i th row of A ,

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + a_i dW(t) = \mu_i dt + \sum_{j=1}^m A_{ij} dW_j(t).$$

Simulation and payoffs

With independent $Z_t = (Z_{k1}, \dots, Z_{kd}) \sim N(0, I)$,

$$S_i(t_{k+1}) = S_i(t_k) \exp \left(\left[\mu_i - \frac{1}{2} \sigma_i^2 \right] \Delta t_k + \sqrt{\Delta t_k} \sum_{j=1}^d A_{ij} Z_{kj+1, j} \right).$$

Under the risk-neutral measure, $\mu_i = r - \delta_i$. Typical multi-asset payoffs:

Spread option:

$$\mathbb{E}[|S_1(T) - S_2(T)| - K]^+.$$

Basket option:

$$([c_1 S_1(T) + c_2 S_2(T) + \dots + c_d S_d(T)] - K)^+.$$

Outperformance option:

$$(\max(c_1 S_1(T), c_2 S_2(T), \dots, c_d S_d(T)) - K)^+.$$

Two-asset barrier example:

$$\mathbf{1}_{[\min_{i=1, \dots, n} S_2(t_i) < b]} (K - S_1(T))^+.$$

4 Variance Reduction Techniques

4.1 Control Variates

Basic method

Let $Y_1, \dots, Y_b$ be i.i.d. outputs for estimating $\mathbb{E}[Y]$, and suppose X_i is generated with Y_i and $\mathbb{E}[X]$ is known. For fixed b define

$$Y_i(b) = Y_i - b(X_i - \mathbb{E}[X]),$$

and estimate

$$\bar{Y}(b) = \bar{Y} - b(\bar{X} - \mathbb{E}[X]) = \frac{1}{b} \sum_{i=1}^b [Y_i - b(X_i - \mathbb{E}[X])].$$

This is unbiased for all b . Its per-replication variance is

$$\sigma^2(b) = \text{var}[Y_i(b)] = \sigma_Y^2 - 2b\sigma_X \rho_{XY} + b^2 \sigma_X^2.$$

The optimal coefficient is

$$b^* = \frac{\sigma_Y}{\sigma_X} \rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{var}(X)}.$$

The resulting variance ratio relative to plain Monte Carlo is

$$\frac{\text{var}[\bar{Y} - b^*(\bar{X} - \mathbb{E}[X])]}{\text{var}[\bar{Y}]} = 1 - \rho_{XY}^2.$$

Thus speed-up is approximately $1/(1 - \rho_{XY}^2)$. The sign of ρ_{XY} is absorbed by b^* ; only $|\rho_{XY}|$ matters.

Estimated coefficient and CI

In practice use the sample regression coefficient

$$\hat{b}_n = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Then $\hat{b}_n \rightarrow b^*$ a.s. and

$$\bar{Y}(\hat{b}_n) = \frac{1}{b} \sum_{i=1}^n \{Y_i - \hat{b}_n(X_i - \mathbb{E}[X])\}.$$

Using $\hat{b}_n$ creates finite-sample bias, but has the same asymptotic precision as using b^* . For fixed b ,

$$\frac{\bar{Y}(b) - \mathbb{E}[\bar{Y}]}{\sigma(b)/\sqrt{m}} \Rightarrow N(0, 1),$$

with sample standard deviation

$$s(b) = \left[\frac{1}{n-1} \sum_{i=1}^n (Y_i(b) - \bar{Y}(b))^2 \right]^{1/2}.$$

Hence an asymptotic CI is $\bar{Y}(b) \pm z_{\delta/2} s(b)/\sqrt{n}$. Also

$$\frac{\bar{Y}(\hat{b}_n) - \mathbb{E}[\bar{Y}]}{s(\hat{b}_n)/\sqrt{n}} \Rightarrow N(0, 1).$$

Financial controls

Any martingale with known initial value is a natural control. Under risk-neutral pricing with constant r , $e^{-rT} S(t)$ is a martingale, so $\mathbb{E}[e^{-rT} S(T)] = S(0)$. For discounted payoff Y ,

$$\frac{1}{b} \sum_{i=1}^n \{Y_i - b[e^{-rT} S_i(T) - S(0)]\}$$

uses the stock as control. For call, this works best near $K = 0$ and may weaken for deep OTM strikes. A stronger control can be a tractable option. For Asian options,

$$\bar{S}_A = \frac{1}{n} \sum_{i=1}^n S(t_i), \quad \bar{S}_G = \left(\prod_{i=1}^n S(t_i) \right)^{1/n}.$$

Under GBM, $\bar{S}_G$ is lognormal and the geometric-average Asian option can be priced analytically; its payoff is often highly correlated with the arithmetic-average Asian payoff.

4.2 Antithetic Variates

Estimator

Antithetic variates pair inputs to create negative dependence while preserving the marginal law. If $U \sim \text{Unif}[0, 1]$, use U and $1 - U$; for a symmetric normal input, use Z and $-Z$.

Let $(Y_i, \bar{Y}_i)$ be i.i.d. pairs with the same marginal distribution. The estimator is

$$Y_{\mathcal{A}\mathcal{V}} = \frac{1}{n} \sum_{i=1}^n Y_i + \sum_{i=1}^n \bar{Y}_i = \frac{1}{n} \sum_{i=1}^n \frac{Y_i + \bar{Y}_i}{2}.$$

Apply the CLT to pair averages:

$$\frac{Y_{\mathcal{A}\mathcal{V}} - \mathbb{E}[Y]}{\sigma_{\mathcal{A}\mathcal{V}}/\sqrt{n}} \Rightarrow N(0, 1), \quad \sigma_{\mathcal{A}\mathcal{V}}^2 = \text{var} \left(\frac{Y_1 + \bar{Y}_1}{2} \right).$$

A CI is $Y_{\mathcal{A}\mathcal{V}} \pm z_{\delta/2} s_{\mathcal{A}\mathcal{V}}/\sqrt{n}$, where $s_{\mathcal{A}\mathcal{V}}$ is the sample sd of pair averages.

When it helps

Compare n antithetic pairs with $2n$ independent samples. Since

$$\text{var}[Y_1 + \bar{Y}_1] = 2 \text{var}[Y_1] + 2 \text{Cov}(Y_1, \bar{Y}_1),$$

antithetic sampling reduces variance iff $\text{Cov}(Y_1, \bar{Y}_1) < 0$. Monotone outputs often satisfy this when paired by $U, 1 - U$ or $Z, -Z$.

GBM call example

For a call, pair Z_t and $-Z_t$:

$$S_t^+ = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z_t \right),$$

$$S_t^- = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T - \sigma \sqrt{T} Z_t \right).$$

Set $X_t = e^{-rT} (S_t^+ - K)^+$ and $Y_t = e^{-rT} (S_t^- - K)^+$. Then

$$\hat{v} = \frac{1}{n} \sum_{i=1}^n \frac{X_i + Y_i}{2},$$

$$SE = \left[\frac{1}{n(n-1)} \prod_{i=1}^n \left(\frac{X_i + Y_i}{2} \right)^2 - n \hat{v}^2 \right]^{1/2}.$$

4.3 Stratified Sampling

Basic method

Let $A_1, \dots, A_K$ be disjoint strata with probabilities p_i . Then

$$\mathbb{E}[Y] = \sum_{i=1}^K p_i \mathbb{E}[Y \mid Y \in A_i].$$

With proportional allocation $n_i = np_i$ and conditional draws $Y_{ij} \sim (Y \mid Y \in A_i)$,

$$\hat{Y} = \sum_{i=1}^K p_i \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^K \sum_{j=1}^{n_i} Y_{ij}.$$

Stratification removes the random variation in the number of observations landing in each stratum.

General allocation

Often strata are defined by another variable X . If A_i partition the support of X ,

$$\mathbb{E}[Y] = \sum_{i=1}^K p_i \mathbb{E}[Y \mid X \in A_i].$$

For $n_i \geq 1$, $\sum_i n_i = n$, and $q_i = n_i/n$,

$$\hat{Y} = \sum_{i=1}^K p_i \frac{n_i}{\hat{n}_i} \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^K \frac{p_i}{q_i} \sum_{j=1}^{n_i} Y_{ij}.$$

Let $\mu_i = \mathbb{E}[Y \mid X \in A_i]$ and $\sigma_i^2 = \text{var}(Y \mid X \in A_i)$. Then $\hat{Y}$ is unbiased and

$$\text{var}[\hat{Y}] = \sum_{i=1}^K p_i \frac{\sigma_i^2}{q_i} = \frac{\sigma^2(q)}{n}, \quad \sigma^2(q) = \sum_{i=1}^K p_i \frac{\sigma_i^2}{q_i}.$$

With fixed K , $\sqrt{n}(\hat{Y} - \mu) \Rightarrow N(0, \sigma^2(q))$, and

$$s^2(q) = \sum_{i=1}^K p_i \frac{\hat{\sigma}_i^2}{q_i} \hat{\sigma}_i^2$$

estimates $\sigma^2(q)$.

Uniform and inverse-transform strata

For $U \sim \text{Unif}[0, 1]$, equal strata are $A_i = ((i-1)/n, i/n]$. With $U_i \sim \text{Unif}[0, 1]$,

$$V_i = \frac{i-1}{n} + \frac{U_i}{n}, \quad Y = \frac{1}{n} \sum_{i=1}^n f(V_i).$$

For a distribution with CDF F , choose probabilities p_i and quantiles $a_i = F^{-1}(p_1 + \dots + p_i)$. Conditional simulation in stratum $(a_{i-1}, a_i]$ uses

$$V = F(a_{i-1}) + U \{F(a_i) - F(a_{i-1})\}, \quad F^{-1}(V) \sim (Y \mid Y \in A_i).$$

Proportional and optimal allocation

For proportional allocation $q_i = p_i$,

$$\sigma^2(q) = \sum_{i=1}^K p_i \sigma_i^2.$$

Plain variance decomposes as

$$\text{var}(Y) = \sum_{i=1}^K p_i \sigma_i^2 + \sum_{i=1}^K p_i \mu_i^2 - \left(\sum_{i=1}^K p_i \mu_i \right)^2.$$

By Jensen, proportional stratification cannot increase variance, with strict gain unless all μ_i are equal. Optimizing $\sigma^2(q)$ gives

$$q_i^* = \frac{p_i \sigma_i}{\sum_{i=1}^K p_i \sigma_i}, \quad \sigma^2(q^*) = \left(\sum_{i=1}^K p_i \sigma_i \right)^2.$$

Thus allocate samples proportional to stratum probability times within-stratum sd.

Call option strata

For a GBM call, write $W_t = \sqrt{T} \Phi^{-1}(U)$ with $U \sim \text{Unif}[0, 1]$ and

$$h(u) = \left(S_0 \exp \left\{ -\frac{1}{2} \sigma^2 T + \sigma \sqrt{T} \Phi^{-1}(u) \right\} - e^{-rT} K \right)^+.$$

With k equal strata, $U_{ij} = (i-1)/k + V_{ij}/k$, $\hat{\mu}_i = n_i^{-1} \sum_{j=1}^{n_i} h(U_{ij})$, and

$$\hat{\mu} = \frac{1}{k} \sum_{i=1}^k \hat{\mu}_i, \quad SE = \frac{1}{k} \left(\sum_{i=1}^k \frac{\hat{\sigma}_i^2}{n_i} \right)^{1/2}.$$

4.4 Importance Sampling

Change of measure

To estimate $\mu = \mathbb{E}[h(X)]$ with density f , choose an alternative density g such that $g(x) = 0$ implies $f(x) = 0$. If $Y \sim g$,

$$\mu = \int h(x) f(x) dx = \int h(x) \frac{f(x)}{g(x)} g(x) dx = \mathbb{E}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right].$$

The likelihood ratio is $f(Y)/g(Y)$, and the estimator averages $h(Y_i) f(Y_i)/g(Y_i)$.

In a discrete model, use $h(Y_i) p(Y_i)/\bar{p}(Y_i)$.

Variance and zero-variance density

The IS estimator has variance

$$\frac{1}{n} \text{var}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right].$$

For $h \geq 0$, the zero-variance density is

$$g^*(x) = c h(x) f(x), \quad \frac{1}{c} = \int h(x) f(x) dx = \mu.$$

It is unusable directly because it requires μ , but it says g should mimic $h(x) f(x)$.

Normal mean shift and options

For $X \sim N(0, 1)$, choose $Y \sim N(\theta, 1)$. The likelihood ratio is

$$\frac{f(x)}{g(x)} = \exp \left\{ -\theta x + \frac{1}{2} \theta^2 \right\}.$$

Mode matching sets $\theta = \arg \max_x h(x) f(x)$. For a binary call,

$$\mathbb{E}[e^{-rT} \mathbf{1}_{\{S_T \geq K\}}] = \mathbb{E}[e^{-rT} \mathbf{1}_{\{X \geq b\}}],$$

where

$$b = \frac{\log(K/S_0)}{\sigma \sqrt{T}} - \frac{(r - \sigma^2/2)T}{\sigma \sqrt{T}}.$$

Here $h(x) = e^{-rT} \mathbf{1}_{\{x \geq b\}}$, $x^* = \max\{b, 0\}$, and with $Y \sim N(x^*, 1)$ the output is

$$h(Y) \frac{f(Y)}{g(Y)} = \begin{cases} 0, & Y < b, \\ \exp\{-rT - x^* Y + (x^*)^2/2\}, & Y \geq b. \end{cases}$$

For a vanilla call,

$$h(x) = \left(S_0 \exp \left\{ -\frac{1}{2} \sigma^2 T + \sigma \sqrt{T} x \right\} - e^{-rT} K \right)^+,$$

and the shift pushes sampling toward exercise states.

Exponential tilting

For density f , the exponential tilt family is

$$f_\theta(x) = \frac{e^{\theta x} f(x)}{\int e^{\theta x} f(x)}, \quad \theta \in \mathbb{R}.$$

For a pmf p ,

$$p_\theta(x) = \frac{e^{\theta x} p(x)}{\sum_x e^{\theta x} p(x)}.$$

For normal X , this is mean shifting. For rare large losses, take $\theta > 0$.

Bernoulli rare event

Let independent draws satisfy $P(X_k = 1) = p_k$ and $L = \sum_{k=1}^m X_k$. To estimate $P(L > x)$ with $x > \mathbb{E}[L]$, tilt each Bernoulli:

$$\bar{p}_k = P(Y_k = 1) = \frac{p_k e^\theta}{1 + p_k(e^\theta - 1)}, \quad \bar{L} = \sum_{k=1}^m Y_k.$$

The IS output is

$$H = \mathbf{1}_{\{\bar{L} > x\}} \prod_{k=1}^m \left(\frac{p_k}{\bar{p}_k} \right)^{Y_k} \left(\frac{1 - p_k}{1 - \bar{p}_k} \right)^{1 - Y_k}.$$

With

$$\phi(\theta) = \sum_{k=1}^m \log[1 + p_k(e^\theta - 1)],$$

one has $\mathbb{E}[H^2] \leq e^{-2\phi(\theta) + 2\phi(0)}$. Choose $\theta > 0$ by solving $\phi'(\theta) = x$.

4.5 Cross-Entropy Method

Principle

Cross-entropy selects a parametric IS density f_θ close to the zero-variance density

$$g^*(x) = \frac{1}{h} h(x) f(x).$$

Minimize

$$R(g^* \| f_\theta) = \int \log \left(\frac{g^*(x)}{f_\theta(x)} \right) g^*(x) dx,$$

equivalently maximize

$$\int h(x) f(x) \log f_\theta(x) dx.$$

The population first-order equation is

$$\mathbb{E} \left[h(X) \frac{\partial}{\partial \theta} \log f_\theta(X) \right] = 0.$$

Basic algorithm

Using pilot samples $X_1, \dots, X_N \sim f$, solve

$$\frac{1}{N} \sum_{k=1}^N h(X_k) \frac{\partial}{\partial \theta} \log f_\theta(X_k) = 0.$$

For $f = N(0, I_m)$ and $f_\theta = N(\theta, I_m)$,

$$\hat{\theta} = \frac{\sum_{k=1}^N h(X_k) X_k}{\sum_{k=1}^N h(X_k)}.$$

Then sample $Y_i \sim f_{\hat{\theta}}$ and average

$$H_i = h(Y_i) \frac{f(Y_i)}{f_{\hat{\theta}}(Y_i)}.$$

For a one-dimensional normal shift, $H_i = h(Y_i) \exp\{-\hat{\theta} Y_i + \hat{\theta}^2/2\}$. For very rare events, the basic method may fail because $\sum_k h(X_k) = 0$ in the pilot run.

Iterative cross-entropy

Let $t_\theta(x) = f(x)/f_\theta(x)$. If pilot samples are drawn from f_v , then maximize

$$\mathbb{E}_v[h(Y) t_v(Y) \log f_\theta(Y)],$$

and solve

$$\mathbb{E}_v \left[h(Y) t_v(Y) \frac{\partial}{\partial \theta} \log f_\theta(Y) \right] = 0.$$

At iteration j , sample $Y_k \sim f_{\hat{\theta}_j}$ and update from

$$\frac{1}{N} \sum_{k=1}^N h(Y_k) t_{\hat{\theta}_j}(Y_k) \frac{\partial}{\partial \theta} \log f_\theta(Y_k) = 0.$$

For $N(\theta, I_m)$,

$$\hat{\theta}_{j+1} = \frac{\sum_{k=1}^N h(Y_k) t_{\hat{\theta}_j}(Y_k) Y_k}{\sum_{k=1}^N h(Y_k) t_{\hat{\theta}_j}(Y_k)}.$$

Here

$$t_{\hat{\theta}_j}(Y_k) = \exp(-(\hat{\theta}_j, Y_k) + \|\hat{\theta}_j\|^2/2).$$

Usually only a few iterations are needed; convergence means the updates oscillate in a small range rather than become deterministic.

5 Discretization Methods

5.1 Euler And Milstein Schemes

Euler scheme

For the SDE $dX(t) = a(X(t)) dt + b(X(t)) dW(t)$, with $X \in \mathbb{R}^d$ and $W \in \mathbb{R}^m$, the Euler approximation on $t_i = ih$ is

$$\hat{X}(i+1) = \hat{X}(i) + a(\hat{X}(i))h + b(\hat{X}(i))\Delta W(i)/\sqrt{\Delta t} Z_{i+1},$$

where Z_i are independent m -dimensional standard normals. Euler freezes the drift and diffusion integrands on each interval. The drift is order h , while the diffusion increment is order $\sqrt{h}$, so a first refinement focuses on the diffusion term.

Scalar Milstein refinement

In the scalar case, Ito's formula gives

$$db(X(t)) = [b'(X(t))a(X(t)) + \frac{1}{2} b''(X(t))b^2(X(t))] dt$$

$$+ b'(X(t))b(X(t))dW(t).$$

Dropping the higher-order drift part,

$$b(X(t)) \approx b(X(t)) + b'(X(t))b(X(t))[W(t) - W(t)].$$

Multivariate normal score

If $X \sim \mathcal{N}(\mu(\theta), \Sigma)$, then

$$\frac{d}{d\theta} \log g_\theta(X) = (X - \mu(\theta))^\top \Sigma^{-1} \dot{\mu}(\theta).$$

If $X = \mu(\theta) + AZ$, $Z \sim \mathcal{N}(0, I)$, and $AA^\top = \Sigma$, the score is

$$\frac{d}{d\theta} \log g_\theta(X) = Z^\top A^{-1} \dot{\mu}(\theta).$$

If θ is in Σ ,

$$\frac{d}{d\theta} \log g_\theta(X) = -\frac{1}{2} \operatorname{tr}(\Sigma^{-1} \dot{\Sigma}) + \frac{1}{2} (X - \mu)^\top \Sigma^{-1} \dot{\Sigma} \Sigma^{-1} (X - \mu).$$

For d-asset GBM with $dS_t(t)/S_t(t) = r dt + v_1^\top dW(t)$, let A have rows $v_1^\top$ and

$$\Sigma = AA^\top.$$

$$X \sim \mathcal{N}(M, T\Sigma), \qquad \mu_i = \log S_i(0) + (r - \frac{1}{2} \|v_i\|_2^2)T.$$

For payoff $f(S_1(T), \dots, S_d(T))$, the LRM delta for asset i is

$$f(S_1(T), \dots, S_d(T)) \frac{(\frac{Z^\top A^{-1})_i}{S_i(0)\sqrt{T}}}.$$

6.4 Gamma And Examples

LRM gamma

If $\dot{g}_\theta$ is the second density derivative, then

$$f(X) \frac{\dot{g}_\theta(X)}{g_\theta(X)}$$

estimates $d^2 \mathbb{E}_\theta [f(X)]/d\theta^2$, under valid derivative-expectation interchanges.

For Black-Scholes gamma,

$$\frac{d^2 \mathbb{E}(S(T)) dS^2(0)}{g(S(T))} = \frac{\mathbb{E}(S(T))^2 - 1}{S(0)^2 \sigma^2 T} - \frac{\mathbb{E}(S(T))}{S(0)^2 \sigma \sqrt{T}}.$$

Replace $\mathbb{E}(S(T))$ by Z when using the usual GBM simulation. Higher derivatives introduce heavier powers and often larger variance. For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$\frac{\dot{g}_\mu(X)}{g_\mu(X)} = \frac{X - \mu}{\sigma^2}, \qquad \frac{\ddot{g}_\mu(X)}{g_\mu(X)} = \frac{(X - \mu)^2 - \sigma^2}{\sigma^4}.$$

Artificial-dynamics control for delta

For $dS_t = rS_t dt + \theta S_t dW_t$, $d\theta_t = a(\Theta - \theta_t) dt + \beta dB_t$, write $S_t = e^{S_t Y_t}$ with

$$dY_t = (r - \frac{1}{2} \theta_t^2) dt + \theta_t dW_t, \qquad Y_0 = 0.$$

Then call delta is

$$\Delta = \mathbb{E} \left[\mathbf{1}_{\{Y_T \geq y\}} e^{Y_T - rT} \right], \qquad y = \log(K/S_0).$$

Use Euler for $\langle Y, \Theta \rangle$ and an artificial constant-volatility process with the same Z_t :

$$\tilde{Y}_{t+1} = \tilde{Y}_t + (r - \frac{1}{2} \sigma^2) \Delta t_i + \sigma \sqrt{\Delta t_i} Z_{t+1}.$$

The control is $V = \mathbf{1}_{\{Y_T \geq y\}} e^{Y_T - rT}$ with

$$\mathbb{E}[V] = \Phi(\alpha), \qquad \alpha = \frac{1}{\sigma \sqrt{T}} \log \left(\frac{S_0}{K} \right) + \left(\frac{r}{\sigma} + \frac{r}{\sigma} \right) \sqrt{T}.$$

Use output $X - b[V - \Phi(\alpha)]$, typically with $\sigma = \Theta$.

Euler sensitivity recursion

For $dS_t = rS_t dt + \sigma S_t dW_t$ and $V(S_0) = \mathbb{E}[e^{-rT} h(S_T)]$,

$$\Delta = \mathbb{E} \left[e^{-rT} h'(S_T) \frac{\partial S_T}{\partial S_0} \right].$$

Euler simulation:

$$\hat{S}_0 = S_0, \qquad \hat{S}_{t+1} = \hat{S}_t + r\hat{S}_t \Delta t_i + \sigma(\hat{S}_t) \sqrt{\Delta t_i} Z_{t+1}.$$

Define $\Delta_t = \partial \hat{S}_t / \partial S_0$. Then

$$\hat{\Delta}_0 = 1, \qquad \hat{\Delta}_{t+1} = \hat{\Delta}_t + r\hat{\Delta}_t \Delta t_i + \sigma'(\hat{S}_t) \hat{\Delta}_t \sqrt{\Delta t_i} Z_{t+1}.$$

Estimate delta by averaging $e^{-rT} h'(\hat{S}_T) \hat{\Delta}_T$.

7 Pricing American Options

7.1 Backward Induction And LSM

American exercise problem

Simulation generates paths forward, but an American option needs a backward exercise rule. For a put with exercise dates $t_1 < \dots < t_n = T$ and $t_{j+1} - t_j = \Delta t$, simulate paths $\{S(t_1), \dots, S(t_n)\}$ and define Y_j as the cash flow from t_j onward under the selected policy.

At maturity,

$$Y_n = \max(K - S(t_n), 0).$$

For earlier dates, the generic cash-flow recursion is

$$Y_j = \begin{cases} K - S(t_j), & \text{exercise,} \\ e^{-r\Delta t} Y_{j+1}, & \text{continue.} \end{cases}$$

There is no exercise right before t_1 , so the American put is priced as a claim paying Y_1 at t_1 :

$$P_A(0, S) = \mathbb{E} \left[e^{-rt_1} Y_1 \mid S_0 = S \right], \qquad P_A(0, S) = \frac{1}{N} \sum_{i=1}^N e^{-rt_1} Y_1^{(i)}.$$

Continuation value

At t_j , the decision must use information available at t_j . Compare immediate exercise $K - S(t_j)$ with the conditional discounted continuation value

$$f_j(S(t_j)) = \hat{\mathbb{E}} \left[e^{-r\Delta t} Y_{j+1} \mid S(t_j) \right].$$

Thus

$$Y_j = \begin{cases} K - S(t_j), & K - S(t_j) \geq f_j(S(t_j)), \\ e^{-r\Delta t} Y_{j+1}, & K - S(t_j) < f_j(S(t_j)). \end{cases}$$

For out-of-the-money put paths, $K - S(t_j) < 0$, set $f_j(S(t_j)) = 0$ and continue:

$$Y_j = e^{-r\Delta t} Y_{j+1}.$$

Least-squares approximation

Longstaff-Schwartz estimates f_j from simulated paths by least squares. Use only in-the-money paths in the regression, because only they are close to the exercise decision. Approximate

$$f_j(S(t_j)) \approx \sum_{k=0}^{\text{var}} a_k S(t_j)^k.$$

Regress the least-squared continuation value $e^{-r\Delta t} Y_{j+1}$ on basis functions of $S(t_j)$. Degree-3 polynomials are often reasonable; Hermite, Laguerre, and Legendre bases are common alternatives, and prices are usually fairly robust to the basis choice.

7.2 Three-Date American Put Example

Setup

Let $S(t)$ follow risk-neutral GBM with $S(0) = 10$, $r = 3\%$, dividend yield $\delta = 0$, $\sigma = 40\%$, strike $K = 12$, maturity $T = 1$, and exercise dates $1/3, 2/3, 1$. Simulate with $S(t + \Delta t) = S(t) \exp \left[(r - \frac{\sigma^2}{2}) \Delta t + \sigma \Delta W(t) \right]$.

Using the 8 sample paths in the lecture, the terminal payoff $Y_3 = \max(K - S(1), 0)$ gives the European put estimate 2.4343, a lower bound for the American put.

Step at $t = 2/3$

For in-the-money paths, regress $Y_3 e^{-r\Delta t}$ on $S(2/3)$ using a quadratic:

$$Y_3 e^{-r\Delta t} = \hat{a}_0 + \hat{a}_1 S(2/3) + \hat{a}_2 [S(2/3)]^2 + e.$$

The fitted continuation value is

$$f_2(S) = -82.5347 + 17.7788 S(2/3) - 0.9063 [S(2/3)]^2.$$

Then

$$Y_2 = \begin{cases} K - S(2/3), & K - S(2/3) \geq f_2(S(2/3)), \\ e^{-r\Delta t} Y_3, & \text{otherwise.} \end{cases}$$

Out-of-the-money paths continue, so $Y_2 = e^{-r\Delta t} Y_3$ for those paths.

Step at $t = 1/3$

Repeat the regression using $Y_2 e^{-r\Delta t}$ and $S(1/3)$. The fitted continuation value is

$$f_1(S) = -8.9488 + 3.3104 S(1/3) - 0.2036 [S(1/3)]^2.$$

Use

$$Y_1 = \begin{cases} K - S(1/3), & K - S(1/3) \geq f_1(S(1/3)), \\ e^{-r\Delta t} Y_2, & \text{otherwise.} \end{cases}$$

Finally estimate the current price by averaging $e^{-r\Delta t} Y_1$. In the lecture example, $\hat{P}_3 = 3.0919$, which is above the European estimate 2.4343.

8 Multi-Period Investment Models

8.1 Multi-Period Investment

There are n periods of investment. Denote the beginning and end of j -th period as t_{j-1} and t_j , $j = 1, \dots, n$. There is a risk-free asset and m risky stocks in the market. The net return rate of investing in the risk-free asset in the j -th period is a constant $r_{f,j}$. The excess return rate of investing in stock i in period j is $R_{i,j}$.

Assume that $R_j := (R_{1,j}, \dots, R_{m,j})^\top$, $j = 1, \dots, n$ are independent.

Consider an agent with initial wealth x_0 at the beginning of first period. The agent's decision is the allocation of his money in each period.

8.2 Wealth Dynamics

Denote as X_j the wealth of the agent at time t_j , $j = 0, 1, \dots, n$. Suppose that except for the initial investment, the agent does not inject new money nor withdraw money from the investment account before the terminal time $T := t_n$.

At the beginning of each period j , the agent decides to invest $\pi_{i,j-1}$ dollars in stock i , $i = 1, \dots, m$, and the remaining amount of money in the risk-free asset. Wealth dynamics:

$$X_j = (1 + r_{f,j}) X_{j-1} + \sum_{i=1}^m \pi_{i,j-1} R_{i,j}, \quad j = 1, \dots, N$$

8.3 Expected Utility Maximization

Assume that the agent maximizes the expected utility of her terminal wealth,

$$\max_{\pi_{j-1}, j=1, \dots, N} \mathbb{E}[u(X_N)]$$

where u is the agent's utility function and $\pi_{j-1} = (\pi_{1,j-1}, \dots, \pi_{m,j-1})^\top$, $j = 1, \dots, N$ are the agent's portfolio strategy.

8.4 Dynamic Programming

We apply dynamic programming to solve the problem. Principle of Optimality:

Starting from any point on the optimal path, the remaining decisions must constitute an optimal policy with regard to the point.

Consider time t_k when the agent's wealth then is x . Consider the following problem:

$$\max_{\pi_{k-1}, j=k+1, \dots, n} \mathbb{E}_k[u(X_N) | X_k = x]$$

Denote the above optimal value function as $v_k(x)$.

Dynamic Programming (DP) Equation:

$$v_k(x) = \max_{\pi_k} \mathbb{E}_k[v_{k+1}(X_{k+1}) | X_k = x], \quad k = 0, 1, \dots, N-1$$

where $v_N(x) = u(x)$. Instead of solving a dynamic portfolio selection problem, we solve a family of single-period portfolio selection problem. $(\pi_0^*, \dots, \pi_{N-1}^*)$ is optimal to the dynamic portfolio selection problem if and only if π_k^* is optimal to the above single-period portfolio selection problem, $k = 0, \dots, N-1$.

8.5 Power Utility

In the following, we assume power utility function:

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad x > 0$$

for some $0 < \gamma \neq 1$. Applying DP, we have:

$$v_{N-1}(x) = \max_{\pi_{N-1}} \mathbb{E}_{N-1}[u(X_N) | X_{N-1} = x] = b_{N-1} x^{1-\gamma}, \quad x > 0$$

where

$$b_{N-1} = \max_{\pi_N} \frac{\mathbb{E}[(1+r_f+w^\top R_N)^{1-\gamma}]}{v_N(x)}$$

Denote the optimizer in the right-hand side of the above as w_{N-1}^* .

By induction, we conclude:

$$v_{k-1}(x) = (\prod_{j=k}^{N-1} (b_j(1-\gamma))) \frac{x^{1-\gamma}}{1-\gamma}$$

where

$$b_{k-1} = \max_w \frac{\mathbb{E}[(1+r_f+w^\top R_k)^{1-\gamma}]}{1-\gamma}, \quad k = 1, \dots, N$$

Denote the optimizer in the right-hand side of the above as w_{k-1}^* . Then, the optimal portfolio is:

$$\pi_{k-1}^* = w_{k-1}^* X_{k-1}, \quad k = 1, \dots, N$$

Myopic Portfolio

With the power utility function, the investor acts at each stage, based on the optimal policy, as if he were faced with a single-period investment, i.e., as if he were myopic. When the utility function is not a power function or when the stock returns are not independent over time, the above is not necessarily true. For power utility functions, it is implicitly assumed that the wealth must remain positive. Thus, the portfolio π needs to be constrained, i.e., $\pi_{i,k} \geq 0$, $i = 1, \dots, n$, and $\sum_{i=1}^m \pi_{i,k} \leq X_k$.

Example: Expected Utility Maximization

Suppose the risk-free net return rate in each period is 0.03, and there is one stock in the market only with i.i.d. excess return rates over time. Suppose that the log return rate of the stock, $\ln(1 + r_f + R_k)$ follows normal distribution with mean 0.07 and variance 0.04. Suppose that the investor is not allowed to short-sell the stock or to borrow, i.e., $0 \leq R_k \leq X_k$. Suppose that the investor has a power utility function with $\gamma = 10$. To find the optimal portfolio, we need to solve:

$$\max_{w \in [0,1]} f(w)$$

where

$$f(w) := \frac{\mathbb{E}[(1+r_f+R_k w)^{1-\gamma}]}{1-\gamma}$$

Straightforward calculation yields:

$$f'(w) = \mathbb{E}[(1+r_f+R_k w)^{-\gamma} R_k]$$

By binary search, one can find that $f'(0.1489) = 0$, which implies that the optimal portfolio $w^* = 0.1489$, i.e., it is optimal to invest 14.89

8.6 Long-Run Mean Exponential Growth Rate

Suppose $t_k - t_{k-1} = \Delta$, $k = 1, \dots, N$ in the above, e.g., Δ is one month, so the terminal time $t_N = N\Delta$. Assume that the stock excess returns R_k , $k = 1, 2, \dots$ are i.i.d.

For each portfolio strategy $(\pi_1, \pi_2, \dots)$, consider the long-run mean exponential growth rate:

$$\lim_{N \rightarrow +\infty} \mathbb{E} \left[\frac{\log(X_N/X_0)}{N} \right]$$

The objective is to maximize the above long-run mean exponential growth rate.

Kelly's Portfolio

It turns out that the optimal strategy is as follows: at each time $k\Delta$ invest w_i^* proportion of wealth at that time in stock i , $i = 1, \dots, n$, where

$$w^* := (w_1^*, \dots, w_n^*)^\top = \operatorname{argmax}_w \mathbb{E}[\log(1 + r_f + w^\top R_k)]$$

The above strategy is called Kelly strategy.

Kelly portfolio has many interesting properties: Asymptotically maximizes the rate of asset growth. Asymptotically maximizes the median of wealth. Is a myopic strategy.

The Kelly strategy may lead to a large fraction of wealth in bets or in risky investment. For instance, for a stock with annual mean excess return rate 4% and volatility 20%, the Kelly strategy is to invest 100% of wealth in the stock.

Fractional Kelly

MacLean et al. (1992) propose fractional Kelly to tradeoff growth and security. κ -fractional Kelly refers to a strategy of investing κ fraction of one's wealth in the Kelly portfolio and holding the remaining as cash (or in the risk-free asset). Fractional Kelly is a heuristic portfolio, rather than the consequence of optimality. MacLean et al. (1992) lay out two attractive properties of fractional Kelly: It is easily computable. There is an effective tradeoff between growth and security by varying κ .

8.7 Multi-Period Mean-Variance Portfolio

Next, we consider multi-period mean-variance portfolio selection. The market setting is as before. Denote by μ_k and C_k the mean and variance of R_k , respectively.

At time 0, given initial wealth x_0 , the agent considers the following dynamic portfolio selection problem:

$$\begin{aligned} \min_{\pi_0, \dots, \pi_{N-1}} \quad & \frac{1}{2} \operatorname{Var}(X_N) \\ \text{s.t.} \quad & \mathbb{E}[X_N] = z_0 \\ X_j = (1 + r_{f,j}) X_{j-1} + \pi_{j-1}^\top R_j, \quad & j = 1, \dots, N \\ X_0 = x_0 \end{aligned}$$

where z_0 is the agent's target for expected terminal wealth.

Expected Quadratic Utility

Assume that C_k is positive definite, $k = 1, \dots, N$.

Assume that $z_0 > \prod_{j=1}^N (1 + r_{f,j})$.

The mean-variance portfolio selection problem is equivalent to the following:

$$\begin{aligned} \min_{\pi_0, \dots, \pi_{N-1}} \quad & \frac{1}{2} \mathbb{E}[(X_N - z_0)^2] \\ \text{s.t.} \quad & \mathbb{E}[X_N] = z_0 \\ X_j = (1 + r_{f,j}) X_{j-1} + \pi_{j-1}^\top R_j, \quad & j = 1, \dots, N \\ X_0 = x_0 \end{aligned}$$

Lagrange Dual Method

Apply the Lagrange multiplier λ_0 to the constraint for the expected terminal wealth target, and we obtain:

$$\begin{aligned} \min_{\pi_0, \dots, \pi_{N-1}} \quad & \frac{1}{2} \mathbb{E}[(X_N - z_0)^2] - \lambda_0 (\mathbb{E}[X_N] - z_0) \\ \text{subject to} \quad & X_j = (1 + r_{f,j}) X_{j-1} + \pi_{j-1}^\top R_j, \quad j = 1, \dots, N \\ & X_0 = x_0 \end{aligned}$$

Then, we apply dynamic programming to solve the above problem: for each $k = 0, 1, \dots, N-1$ and each x , consider:

$$\begin{aligned} \min_{\pi_k, \dots, \pi_{N-1}} \quad & \frac{1}{2} \mathbb{E}_k[(X_N - z_0)^2 | X_k = x] - \lambda_0 (\mathbb{E}_k[X_N | X_k = x] - z_0) \\ \text{s.t.} \quad & X_j = (1 + r_{f,j}) X_{j-1} + \pi_{j-1}^\top R_j, \quad j = k+1, \dots, N \\ & X_k = x \end{aligned}$$

Denote the optimal value of the above as $v_k(x)$.

Dynamic Programming

DP principle:

$$\begin{aligned} v_k(x) = \min_{\pi_k} \quad & \mathbb{E}_k[v_{k+1}((1 + r_{f,k+1}) X_k + \pi_k^\top \overline{R}_{k+1}) | X_k = x] \\ = \min_{\pi_k} \quad & \mathbb{E}_k[v_{k+1}((1 + r_{f,k+1}) x + \pi_k^\top \overline{R}_{k+1})], \quad k = 0, 1, \dots, N-1 \\ v_N(x) = \quad & \frac{1}{2} x^2 - (z_0 + \lambda_0) x + \frac{1}{2} z_0^2 + \lambda_0 z_0 \end{aligned}$$

Denote $D_k := \mathbb{E}[R_k R_k^\top] = C_k + \mu_k \mu_k^\top$, $k = 1, \dots, N$.

Consequently, the optimal portfolio is:

$$\pi_k^*(x) = (1 + r_{f,k+1}) [(\prod_{j=k}^{N-1} (1 + r_{f,j+1}))^{-1} (z_0 + \lambda_0) - x] D_{k+1}^{-1} \mu_{k+1}$$

where the multiplier λ_0 is to be determined. By maximizing $v_0(x_0)$ with respect to the multiplier λ_0 , we can derive the multiplier:

$$\lambda_0 = -z_0 + \frac{1}{1 - \prod_{j=0}^{N-1} (1 - \frac{\mu_{j+1}^\top D_{j+1}^{-1} \mu_{j+1}}{1 - \beta_{j+1}})} \times [z_0 - (\prod_{j=0}^{N-1} (1 + r_{f,j+1})) (\prod_{j=0}^{N-1} (1 - \frac{\mu_{j+1}^\top D_{j+1}^{-1} \mu_{j+1}}{1 - \beta_{j+1}})) x_0]$$

To summarize, the optimal portfolio is:

(b) If he wants to buy a car insurance which can fully cover his loss in both cases, i.e., stolen and damaged, what is the maximum amount of money he is willing to pay for the insurance?

Solution 3

(a) The Arrow-Pratt absolute risk aversion coefficient is:

$$r(x) = -\frac{u''(x)}{u'(x)} = -\frac{2 \times (1/2) \times (-1/2) x^{-3/2}}{2 \times (1/2) x^{-1/2}} = \frac{1}{2} x^{-1}$$

(b) The maximum amount of money, A, David is willing to pay is the amount such that David is indifferent between his wealth if he purchases the insurance contract and his wealth if he does not purchase the contract.

If David purchases the insurance contract, his wealth is $40000 - A$ and the utility of the wealth is $2\sqrt{40000 - A}$.

If David does not purchase the insurance contract, the expected utility of his wealth is:

$$2\sqrt{40000 - 15000} \times 0.05 + 2\sqrt{40000 - 10000} \times 0.10 + 2\sqrt{40000} \times 0.85 = 390.4524$$

Setting the expected utility of the wealth in the case in which David purchases the insurance equal to the case in which he does not purchase the insurance, namely:

$$2\sqrt{40000 - A} = 390.4524$$

We derive $A = 1886.731$.

Question 4

Consider the following Limit Order Book (LOB) for a security:

Bid Side (Depth Available at Price): \$1.99 (Depth: 2), \$1.98 (Depth: 3), \$1.97 (Depth: 1), \$1.96 (Depth: 2), \$1.95 (Depth: 4).

Ask Side (Depth Available at Price): \$2.02 (Depth: 1), \$2.03 (Depth: 2), \$2.04 (Depth: 2), \$2.05 (Depth: 3), \$2.06 (Depth: 4).

Current Bid Price: \$1.99. Current Ask Price: \$2.02. Current Mid Price: \$2.005.

(a) Suppose an order arrives to buy 3 units of the security at price \$2.00. Calculate the mid price immediately after the arrival of the order.

(b) Suppose an order arrives to sell 1 unit of the security at price \$2.02. Calculate the mid price immediately after the arrival of the order.

(c) Suppose an order arrives to buy 3 units of the security at the market price. Calculate the mid price immediately after the arrival of the order.

(d) Suppose cancellation of an existing sell orders of size 1 at price \$2.02 arrives. Calculate the mid price immediately after the arrival of cancellation.

Solution 4

(a) The order is a buy limit order at \$2.00. Since $2.00 < \text{Ask } (\$2.02)$, it becomes the new best bid. New Bid: \$2.00. Ask remains: \$2.02. Mid Price:

$$\frac{2.02 + 2.00}{2} = \$2.01$$

(b) The order is a sell limit order at \$2.02. Since $2.02 > \text{Bid } (\$1.99)$ and \$2.02 is the current Ask, it adds depth to the ask side but does not change the best prices. Bid: \$1.99. Ask: \$2.02. Mid Price remains:

$$\frac{2.02 + 1.99}{2} = \$2.005$$

(c) A market buy order for 3 units will walk up the book. It consumes the 1 unit at \$2.02. It consumes 2 units at \$2.03. The new best Ask becomes \$2.04. Bid remains \$1.99. Mid Price:

$$\frac{2.04 + 1.99}{2} = \$2.015$$

(d) Cancellation of 1 unit at \$2.02. The existing depth at \$2.02 was 1. After cancellation, depth at \$2.02 becomes 0. The next available ask is \$2.03. New Ask: \$2.03. Bid remains \$1.99. Mid Price:

$$\frac{2.03 + 1.99}{2} = \$2.01$$

Question 5

Consider a 3-period mean-variance portfolio selection problem with one risk-free asset and 1 risky stock. Suppose the initial wealth is \$1. Suppose that the risk-free net return in each period is $r_{f,k} = 0.03$. In addition, for $k = 1, \dots, 3$ the expected excess return vector of the stocks is:

$$\mu_k = \mathbb{E}[R_k] = 5\%$$

and variance of the stock excess return rate is:

$$C_k = \text{Var}(R_k) = 0.09$$

Suppose the agent's target is to grow the wealth by 6% in each period. Find the optimal portfolio strategy of the agent.

Solution 5

The number of investment periods is $N = 3$. We first compute D_k :

$$D_k = \mathbb{E}[R_k R_k^\top] = C_k + \mu_k \mu_k^\top = (5\%)^2 + 0.09 = 0.0925$$

$$\mu_k^\top D_k^{-1} \mu_k = (5\%)^2 / 0.0925 = 0.027027028, \quad k = 1, \dots, 3$$

Next, we compute:

$$D_k^{-1} \mu_k = 5\% / 0.0925 = 0.54054054$$

$$s_k = \prod_{j=k}^{N-1} (1 + r_{f,j+1}) = 1.03^{3-k}, \quad k = 0, \dots, 2$$

$$\beta_k = \prod_{j=k}^{N-1} (1 - \mu_{j+1}^\top D_{j+1}^{-1} \mu_{j+1}) = 0.972972973^{3-k}, \quad k = 0, \dots, 2$$

The target expected final wealth is $z_0 = 1 \times (1 + 0.06)^3$. (The problem says "grow the wealth by 6% in each period", implying $z_0 = (1.06)^3$).

The optimal strategy formula is $\pi_k^* = -A_k x_k + B_k$.

We calculate the coefficients:

$$A_k := (1 + r_{f,k+1}) D_{k+1}^{-1} \mu_{k+1} = 1.03 \times 0.54054054 = 0.556756757, \quad k = 0, 1, 2$$

$$B_k := A_k s_k^{-1} \frac{z_0 - \beta_0 s_0 z_0}{1 - \beta_0}$$

Substituting the values ($s_0 = 1$, $z_0 = (1.06)^3$, $s_0 = 1.03^3$, $\beta_0 = 0.97297^3$):

$$B_k = 0.556756757 \times \frac{(1.06)^3 - 0.972972973^3 \times 1.03^3 \times 1}{1 - 0.972972973^3} \times 1.03^{k-3}$$

$$= 1.30187509 \times 1.03^{k-3}$$

Thus, the values for B_k are:

$$B_k \approx \begin{cases} 1.91400139, & k = 0, \\ 1.227142143, & k = 1, \\ 1.263956407, & k = 2. \end{cases}$$

The optimal portfolio strategy is:

$$\pi_k^* = -0.556756757 x_k + B_k, \quad k = 0, 1, 2$$